Balaji Leninrajan

github.com/BalajiLeninrajan | bleninra@uwaterloo.ca | linkedin.com/in/balaji-leninrajan

Education

University of Waterloo 2023 – Present
Bachelor of Software Engineering *Waterloo, ON, Canada*
Coursework: Operating Systems, Compilers, Data Structures, Algorithms, Databases, Computer Architecture

Experience

Shopify | Production Engineering Intern | Edge Infrastructure *September 2025 – December 2025*
- Engineered a high-scale SSL stack with **Ruby on Rails** and **Terraform**; migrated 10+ services from the legacy stack
- Resolved a **critical production outage** caused by 429-error spikes; Implemented kill switches and refactored core retry logic with intelligent backoffs to maintain site reliability
- Recovered **$100,000+ annually** with an optimized maintenance task to **prune 10M+ dead hostnames efficiently**
- Authored operational playbooks and built **automated incident response tools** to find and fix domain and ssl job processing errors across edge infrastructure

Shopify | Software Engineering Intern | Communications Platform Team *January 2025 – April 2025*
- Shipped features for a high-throughput communications platform using **React**, **Ruby on Rails**, and **GraphQL**, orchestrating **4M+ monthly** emails and push notifications across multiple Shopify services and teams
- Designed an approval workflow system for emails using a **polymorphic Rails concern** that empowers users in marketing automation, resulting in an **80% reduction** in faulty campaigns due to human error
- Optimized error monitoring by implementing intelligent Bugsnag filtering and categorization, reducing non-actionable alerts by 70% and improving response time

Waterloo Aerial Robotics Group | Autonomy Lead *January 2024 – Present*
- Led a team of **10+ software developers** that contributed to back-to-back victories at AEAC 2024 and 2025
- Implemented **3D target mapping** by integrating real-time AI detection with **YOLOv11** and stereo vision for precise spatial localization of targets
- Architected a robust **VSLAM pipeline for autonomous drone navigation,** integrating stereo-depth sensing and visual-inertial odometry (VIO) to ensure precise localization in GPS-denied environments

YTY Group | ERP/Web Development Intern *May 2024 – August 2024*
- Trained an employee attendance prediction model using **TensorFlow** and **Pandas** for data preprocessing, improving HR planning capabilities and reducing administrative workload
- Created a **Flutter**-based expense claims **mobile app** with offline functionality and persistent login, reducing submission time and improving accessibility

Projects

Maram | *Rust, Git*
- Built a **Rust-based Git workflow engine** to eliminate context-switching overhead, automating the orchestration of parallel worktree “sets” for concurrent feature development
- Implemented **conflict-aware synchronization logic** using git2-rs to automate commit propagation (cherry-picking) and rebase management across divergent worktree variants
- Designed a **custom state-management layer** that persists Worktree metadata, allowing for recursive root-discovery and seamless CLI interactions across complex project hierarchies

Linguini | *TypeScript, React, Next.js, tRPC, Tailwind, PostgreSQL*
- Shipped a full-stack multiplayer word-guessing platform using **Next.js, tRPC, and PostgreSQL**
- Created a semantic word-validation engine leveraging **PostgreSQL Recursive CTEs** to traverse hierarchical category graphs, enabling complex, multi-level word-to-category matching
- Optimized database performance by implementing **B-tree indexes** and **SQL Views**, reducing latency for high-concurrency queries and game state updates
- Architected a type-safe communication layer with **tRPC**, ensuring seamless state synchronization between a **responsive React frontend** and the backend for a low-latency game loop

ShowerScribe | *Flask, Python, Bootstrap, SQL, HTML/CSS, Raspberry Pi*
- Engineered an **AI-powered voice assistant** on Raspberry Pi for hands-free, intelligent note-taking
- Developed a **Flask web portal** to provide a centralized interface for organizing and playing back recordings
- Scripted a **Bash orchestration suite** to automate service initialization and optimize boot times for rapid startup
- Integrated Cohere LLM services for **automated summarization**, AI brainstorming, and **semantic search**

Technical Skills

Languages: Python, C, C++, Go, Ruby, Rust, TypeScript, JavaScript, SQL
Technologies: Ruby on Rails, React, React Native, Next.js, Django., GraphQL, gRPC, Tailwind, Nginx
Terraform, Git, Linux, AWS, GCP, Raspberry Pi, PyTorch, TensorFlow, Pandas, Docker, Kubernetes